

Multiscale Entanglement Renormalization Ansatz

Review, Formulation, and Simulation of the Transverse-Field Ising Model

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Abstract

The Multiscale Entanglement Renormalization Ansatz (MERA) is a tensor-network framework designed to represent quantum many-body states while preserving correlations across multiple length scales. It is particularly suited to critical systems, where conventional coarse-graining methods can accumulate short-range entanglement and become computationally inefficient.

This project reviews the motivation, construction, causal structure, and computational interpretation of MERA. The discussion begins with entanglement entropy, area laws, real-space renormalization, tree tensor networks, and tensor-network states. MERA is then formulated by combining coarse-graining isometries with disentanglers that remove short-range entanglement before coarse-graining. A Python-based variational MERA simulation is implemented for a one-dimensional transverse-field Ising model with 256 sites. Ground-state energies and scaling dimensions are calculated and compared with reference or analytical expectations.

The study demonstrates how tensor-network geometry controls computational complexity and how the bounded width of MERA causal cones enables efficient evaluation of local observables. The results show convergence of the variational energy and good agreement of the extracted scaling dimensions with the expected critical behavior.

1 Motivation

The Hilbert space of a quantum system grows exponentially with the number of degrees of freedom. Direct exact diagonalization therefore becomes impractical even for moderately large many-body systems. Numerical methods must exploit physical structure, particularly the fact that physically relevant states often occupy only a small, structured region of the full Hilbert space.

Entanglement provides the organizing principle for tensor-network methods. Ground states of many local, gapped systems satisfy an area law for entanglement entropy, while critical systems exhibit characteristic logarithmic violations of the one-dimensional area law. MERA is designed to handle this scale-dependent entanglement by removing short-range correlations before reducing the number of effective degrees of freedom.

2 Objectives

The project had four main objectives:

1. Review the physical and mathematical motivation for tensor-network methods.
2. Explain the construction of real-space renormalization, tree tensor networks, and MERA.
3. Implement a variational MERA calculation for a one-dimensional quantum Ising model.
4. Evaluate the quality and limitations of the approximation through ground-state energies and scaling dimensions.

3 Theoretical background

3.1 Entanglement and reduced density matrices

For a bipartite state $|\psi\rangle_{AB}$, the reduced density matrix of subsystem A is

$$\rho_A = \text{Tr}_B (|\psi\rangle\langle\psi|).$$

The von Neumann entanglement entropy is

$$S_A = -\text{Tr}(\rho_A \log \rho_A).$$

The scaling of S_A with subsystem size determines how efficiently a state can be represented by a tensor network. In one-dimensional gapped systems, the entropy typically saturates with subsystem size. At a critical point, it often grows logarithmically.

3.2 Real-space and tree renormalization

Real-space renormalization replaces a group of microscopic degrees of freedom by a smaller effective description. In a tensor-network formulation, isometries map a larger local Hilbert space to a reduced effective space while approximately preserving the low-energy properties of interest.

A tree tensor network performs repeated coarse-graining but does not remove all short-range entanglement between neighboring blocks. This entanglement can accumulate across renormalization layers and force the effective bond dimension to grow.

3.3 Entanglement renormalization

MERA introduces disentanglers between neighboring blocks. These unitary tensors reorganize and remove short-range entanglement before coarse-graining is applied. The resulting network retains long-range and scale-dependent correlations more efficiently.

3.4 Tensor-network geometry

MERA is built from:

- **Disentanglers**, which are unitary tensors acting across block boundaries.
- **Isometries**, which map several sites to fewer effective sites.
- **A top tensor**, which represents the final coarse-grained degrees of freedom.
- **Ascending superoperators**, which map local operators to coarser layers.
- **Descending superoperators**, which map density matrices toward finer layers.

A local observable depends only on the tensors inside its causal cone. Because the causal-cone width remains bounded, local observables can be evaluated without contracting the full exponentially large state space.

4 Model system

The numerical implementation studies a one-dimensional transverse-field Ising model. In a representative convention, the Hamiltonian can be written as

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x,$$

where J is the nearest-neighbor coupling and h is the transverse magnetic-field strength. The report focuses on a critical parameter regime and a finite system of 256 sites.

The exact sign and normalization convention used in the implementation should be stated in the accompanying code documentation.

5 Numerical method

5.1 Variational ansatz

The ground state is approximated by a MERA with a fixed network architecture and bond dimension. The tensor elements are treated as variational parameters subject to isometric and unitary constraints.

5.2 Local observables and causal cones

Local Hamiltonian terms are evaluated by propagating operators upward through the network using ascending superoperators, or equivalently by propagating density matrices downward using descending superoperators. The causal-cone structure limits the number of tensors involved at every layer.

5.3 Tensor optimization

The optimization proceeds iteratively:

1. Construct the relevant reduced density matrices.
2. Ascend Hamiltonian terms to the required layers.
3. Construct the environment of an isometry or disentangler.
4. Perform a singular-value decomposition of the environment.
5. Update the tensor using the unitary/isometric optimality condition.
6. Sweep through the network layers.
7. Repeat until the energy change satisfies the convergence criterion.

The computational cost depends strongly on the tensor dimensions, bond dimension, network type, and number of optimization iterations.

6 Numerical results

6.1 Ground-state energy convergence

The report shows that the variational energy approaches a stable value as the optimization proceeds. The energy error relative to the exact/reference value decreases rapidly at early iterations and reaches a tolerance of approximately 10^{-5} for the reported critical setup.

6.2 Scaling dimensions

Scaling dimensions are obtained from the eigenvalues of the appropriate MERA scaling or superoperator construction. The reported values show good agreement with the expected critical-Ising scaling dimensions.

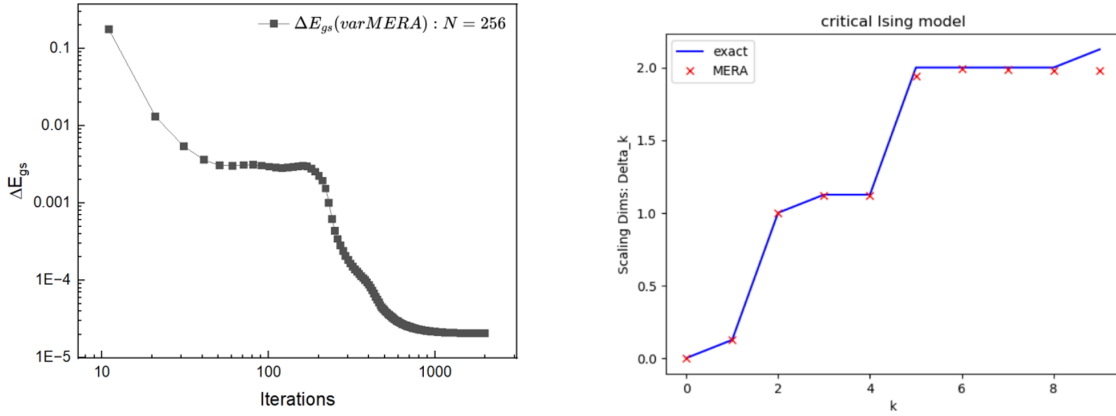


Figure 1: *Relative error: (a) MERA ground-state energy for the 256-site transverse-field Ising model vs optimization iterations, (b) Scaling dimensions extracted from the MERA calculation.*

6.3 Causal-cone diagram

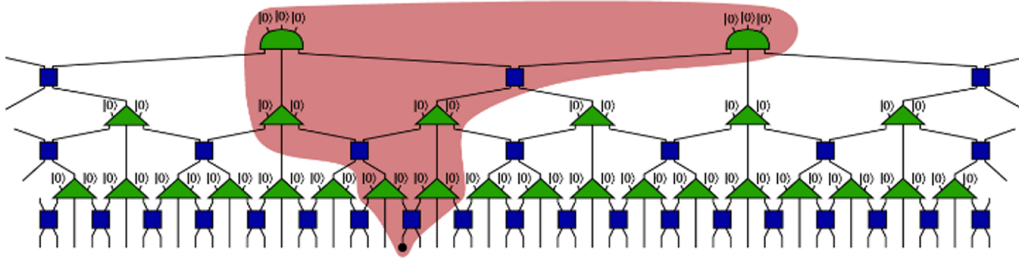


Figure 2: *Bounded causal cone of a local operator in a one-dimensional MERA network. Only tensors inside the cone contribute to the evaluation of the observable; effective site number per level = 2.*

7 Interpretation

The calculation illustrates the central advantage of MERA: the ansatz encodes correlations hierarchically rather than treating the many-body wavefunction as a generic vector in the full Hilbert space. Disentanglers prevent short-range entanglement from accumulating under coarse-graining, while the network geometry naturally organizes correlations by length scale.

The agreement of the extracted scaling dimensions with critical-Ising expectations indicates that the variational network captures important universal information beyond the ground-state energy alone. This is particularly relevant for critical systems, where local observables and scaling behavior are sensitive to the structure of long-range correlations.

8 Limitations and possible improvements

- The accuracy depends on bond dimension, network architecture, tensor initialization, and optimization details.
- The simulation is performed for a finite system rather than directly in the thermodynamic limit.
- The optimization may encounter local minima or slow convergence.
- The implementation does not constitute a systematic benchmark across all MERA variants.
- Reproducibility requires documenting the exact tensor dimensions, conventions, package versions, and update schedule.

Possible extensions include systematic bond-dimension scaling, comparison with MPS/DMRG, finite-size scaling near criticality, observables beyond energy and scaling dimensions, and implementation of symmetry-preserving tensors.

9 Conclusion

This project demonstrates the formulation and practical use of MERA as a tensor-network method for quantum many-body systems. The review establishes why disentanglers are needed, how causal cones control computational complexity, and how variational tensor optimization can be used to approximate ground-state properties. The numerical study of the 256-site transverse-field Ising model produces converged ground-state energies and scaling dimensions consistent with critical behavior. The project therefore combines quantum many-body theory, tensor-network methods, variational optimization, and scientific computing in a single reproducible computational study.

References

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